



INDICADOR DE LOGRO IDL3

Big Data Aplicada

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1 Verificando Hadoop

```
!hadoop version
```

Hadoop 3.4.1

Source code repository <https://github.com/apache/hadoop.git> -r 4d7825309348956336b8f06a08322b78422

Compiled by mthakur on 2024-10-09T14:57Z

Compiled on platform linux-x86_64

Compiled with protoc 3.23.4

From source with checksum 7292fe9dba5e2e44e3a9f763fce3e680

This command was run using /home/hadoop/hadoop/share/hadoop/common/hadoop-common-3.4.1.jar

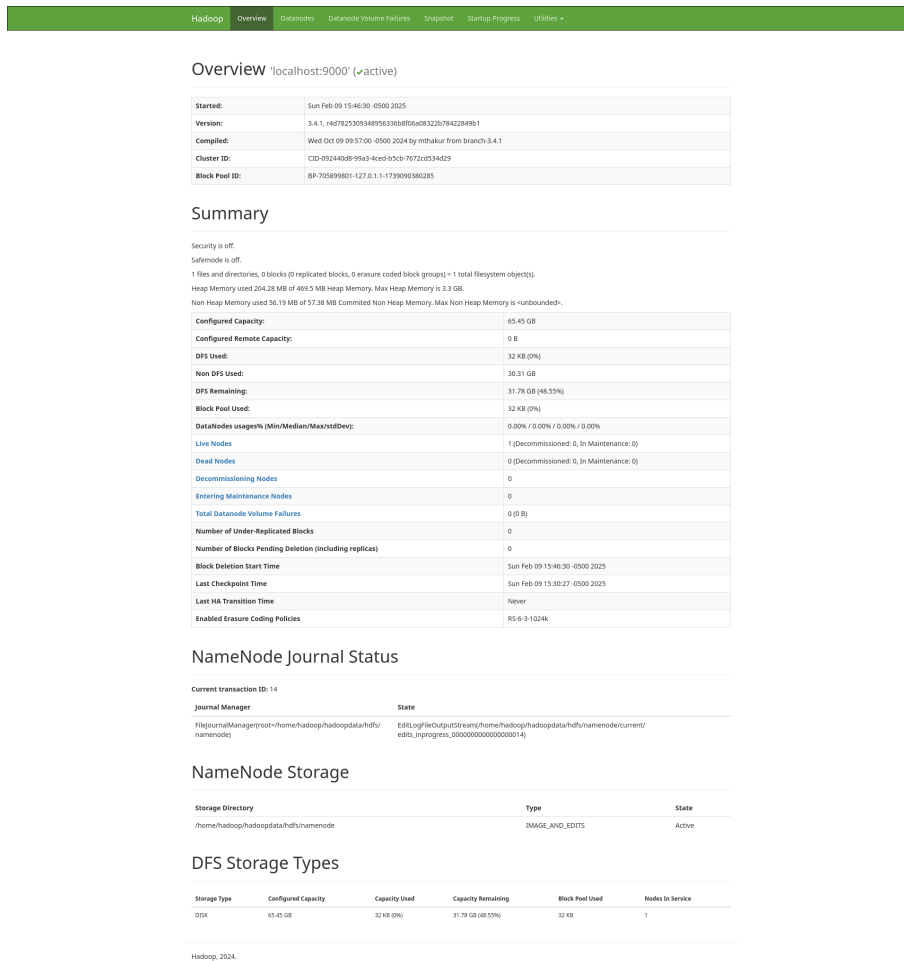


Figura 1: Namenode <http://localhost:9870/>

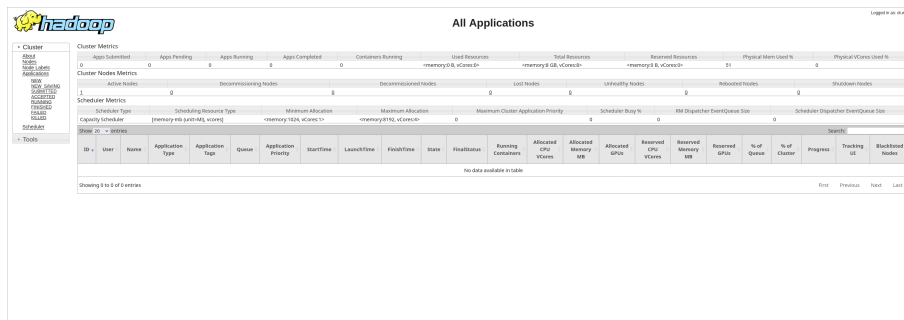


Figura 2: Cluster <http://localhost:8088/>

Hadoop, 2024

```
!spark-shell --version
!start-master.sh
```

Welcome to

version 3.5.4

Revision a6f220d951742f4074b37772485ee0ec7a774e7d

Type --help for more information.

```
starting org.apache.spark.deploy.
```

```
try:
```

```
from pyspark.sql import SparkSession
```

```
# Crear una sesión de Spark
spark = SparkSession.builder \
    .appName("PruebaPySpark") \
    .getOrCreate()
```

```
# Verificar si la sesión está activa
print(f"Versión de Spark: {spark.version}")
print("Sesión de Spark creada con éxito.")
```

```
# Crear un DataFrame simple
datos = [("Juan", 25), ("Ana", 30), ("Pedro", 35)]
columnas = ["Nombre", "Edad"]
df = spark.createDataFrame(datos, columnas)
```

```
# Mostrar el DataFrame
print("Mostrando datos del DataFrame:")
```

```
df.show()

# Contar el número de filas
print(f"Número de filas en el DataFrame: {df.count()}")

# Detener la sesión de Spark
spark.stop()
print("Sesión de Spark detenida.")

except Exception as e:
    print(f"Error al ejecutar el código con PySpark: {e}")
```

```
25/02/09 15:49:46 WARN Utils: Your hostname, ubuntu resolves to a loopback address: 127.0.1.1; use
25/02/09 15:49:46 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
25/02/09 15:49:46 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform...
```

Versión de Spark: 3.5.4

Sesión de Spark creada con éxito.

Mostrando datos del DataFrame:

[Stage 0:> (0 + 1) / 1]

```
+-----+-----+
|Nombre|Edad|
+-----+-----+
|  Juan|  25|
|   Ana|  30|
| Pedro|  35|
+-----+-----+
```

Número de filas en el DataFrame: 3

Sesión de Spark detenida.